

The Accessibility Paradox: When LLM Simplification for Students with Learning Difficulties Amplifies Strategic Misconceptions in Computer Science Education

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Abstract

Large language models (LLMs) are increasingly used as assistive technology by students with reading and attention difficulties (e.g., dyslexia, ADHD), who rely on them as patient, private tutors capable of simplifying complex texts. However, it remains an open question whether this simplification compromises conceptual accuracy. This paper empirically investigates the accessibility paradox: the hypothesis that LLMs' attempts to be inclusive and simple result in an increased rate of strategic misconceptions - subtle, conceptually flawed, yet persuasively formulated responses that are particularly dangerous for vulnerable learners who already expend significant cognitive effort on text decoding and have reduced capacity for parallel fact-verification. We introduce AccessTrap-CS, a benchmark corpus of 735 responses generated by seven open-weight LLMs across 35 core computer science concepts spanning five subfields (7 per subfield). Each concept was prompted under three conditions: (a) a standard explanation, (b) a simplified explanation simulating interaction with a student with reading difficulties, and (c) a highly simplified explanation simulating interaction with a student with severe attention difficulties. Responses were evaluated for strategic misconception rate (SMR), dangerous analogy rate (DAR), and oversimplification error rate (OER) - a novel error category capturing inaccuracies that arise exclusively from the simplification process. Statistical comparisons across models and conditions employ one-way ANOVA with Tukey HSD post-hoc correction and chi-square tests. Based on our findings, we propose three accessibility-first design principles for AI tutoring interfaces that visually signal model uncertainty without relying on text-based disclaimers, thereby protecting the very learners these systems aim to serve.

Keywords: large language models; inclusive education; strategic misconceptions; accessibility; dyslexia; ADHD; assistive technology; AI safety; computer science education; oversimplification errors

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1. Introduction

Large language models (LLMs) are rapidly being adopted as assistive technology by students with reading and attention difficulties [1,2]. For learners with dyslexia, who struggle with text decoding, and students with ADHD, who face sustained attention challenges, LLMs offer a compelling value proposition: a patient, private tutor, available on demand, capable of simplifying complex academic texts in infinitely flexible ways [3,4].

Unlike human tutors, LLMs are available around the clock, never judge repeated questions, and can rephrase explanations until something clicks. It is therefore unsurprising that platforms such as ChatGPT and Claude are increasingly used in this capacity, often without institutional oversight or teacher involvement [5,6].

However, this promising picture conceals a potentially dangerous blind spot. The very features that make LLMs attractive for students with learning difficulties, such as simplified language, short sentences, concrete analogies, and the avoidance of technical jargon, may inadvertently introduce subtle but pedagogically significant errors. We term this phenomenon the *accessibility paradox*: the hypothesis that LLMs' well-intentioned attempts to be inclusive and accessible result in an increased rate of what we call strategic misconceptions: subtly flawed yet persuasively formulated responses that are particularly dangerous for vulnerable learners.

We define a *strategic misconception* as an explanation that is conceptually coherent, internally consistent, and often intuitively appealing, but that nonetheless contains a fundamental misunderstanding of the target concept. Unlike random errors or obvious falsehoods, strategic misconceptions are dangerous precisely because of their plausibility: a student who encounters one is not alerted to an error but is instead led to construct a mental model that is systematically wrong [7]. This definition draws on a broader tradition of misconception research in science and mathematics education [8], which consistently shows that intuitive but incorrect prior conceptions are among the hardest to correct through subsequent instruction. For students with dyslexia and ADHD, who already expend significant cognitive effort on text decoding and attention regulation, the capacity for parallel fact-verification is severely reduced [9,10]. These learners are therefore disproportionately vulnerable to accepting LLM-generated simplifications at face value.

The computer science (CS) education context makes this problem particularly acute. CS concepts are often abstract, hierarchical, and resistant to faithful simplification through everyday analogy [11]. A concept such as a semaphore, a deadlock, or virtual memory cannot be accurately captured by a single intuitive metaphor without loss of essential technical content. Yet, under pressure to simplify, LLMs frequently resort to analogies that are intuitively appealing but technically misleading. For example, comparing a semaphore to a traffic light fails to convey atomicity and counting semantics, and describing a deadlock as two people stuck in a doorway omits resource holding and circular wait conditions. These errors are not isolated or accidental; they are systematic products of the simplification process itself. We term this subclass *oversimplification errors* (OER): errors that would not appear in a standard explanation but emerge exclusively as a consequence of accessibility-driven simplification.

This trade-off between warmth and accuracy is not unique to accessibility contexts. Ibrahim et al. [12] demonstrated that fine-tuning LLMs to produce warmer, more empathetic responses consistently increases error rates across model architectures, with warm models showing substantially higher rates of factual inaccuracies and sycophantic validation of incorrect user beliefs, particularly when users expressed emotional vulnerability. Accessibility-driven simplification represents an analogous pressure: models instructed to be maximally clear and engaging may sacrifice technical precision in ways that standard evaluation benchmarks fail to detect.

Prior work has documented the benefits of LLMs for students with disabilities [1,2] and has separately studied misconceptions in CS education [8,11]. However, no study to date has empirically investigated whether accessibility-driven simplification systematically increases the rate of strategic misconceptions in LLM-generated CS explanations. This gap is critical: if simplification amplifies errors, then current assistive technology practices may be inadvertently harming the very population they aim to serve.

In this paper, we present AccessTrap-CS, a benchmark corpus of 735 responses generated by seven open-weight LLMs across 35 core computer science concepts spanning five subfields: algorithms and data structures, programming languages and paradigms, databases, computer networks, and operating systems (7 per subfield). Each concept was prompted under three conditions: (a) a standard explanation suitable for a first-year CS student (baseline), (b) a simplified explanation simulating interaction with a student with reading difficulties (dyslexia condition), and (c) a highly simplified explanation simulating interaction with a student with severe attention difficulties (ADHD condition). Responses were evaluated using an LLM-as-judge approach [12], with Qwen3:235b serving as the expert evaluator, validated through extreme-case qualitative sampling ($n = 30$).

Our contributions are as follows.

First, we provide empirical evidence of the accessibility paradox, demonstrating that accessibility-driven simplification significantly increases readability and analogy frequency ($p < .001$) while systematically introducing conceptual errors, including errors that do not occur under standard prompting conditions.

Second, we introduce a novel error taxonomy: oversimplification errors (OER) as a distinct error category that emerge exclusively as a product of the simplification process and would not appear in technically precise explanations.

Third, we present the first systematic analysis of dangerous analogies in LLM-generated CS content, documenting cases in which commonly generated analogies (e.g., semaphore to traffic light; deadlock to two people in a doorway) misrepresent critical technical properties of CS concepts.

Fourth, we propose three accessibility-first design principles for AI tutoring interfaces that visually signal model uncertainty and analogy status without relying on text-based disclaimers, thereby protecting the very learners these systems aim to serve.

The remainder of this paper is organised as follows. Section 2 reviews related work on LLMs as assistive technology, text simplification, misconceptions in CS education, and the warmth-accuracy trade-off. Section 3 describes our methodology, including concept selection, model selection, prompt design, and analytical approach. Section 4 presents quantitative and qualitative results. Section 5 discusses the implications of our findings. Section 6 proposes three design principles for accessible AI tutoring interfaces. Section 7 acknowledges limitations. Section 8 concludes.

2. Background and Related Work

The accessibility paradox investigated in this paper sits at the intersection of three distinct research streams: (1) LLMs as assistive technology for students with learning difficulties, (2) text simplification and its pedagogical trade-offs, and (3) misconceptions in computer science education. Each stream has developed independently, and their integration, specifically whether simplification for accessibility systematically amplifies conceptual error, remains unexplored.

2.1. LLMs as Assistive Technology for Dyslexia and ADHD

The adoption of LLMs by students with reading and attention difficulties has outpaced empirical investigation of its pedagogical consequences. For learners with dyslexia, who experience phonological processing deficits and slower text decoding [9], LLMs offer a compelling solution: they can rephrase dense academic prose into shorter sentences, replace rare vocabulary with common alternatives, and maintain semantic coherence without the social costs of requesting repeated clarification from human tutors [1]. Similarly, for students with ADHD, whose working memory and sustained attention limitations make

extended technical reading challenging [10], LLMs provide on-demand summarisation and simplification that reduces cognitive load and allows task segmentation [2].

Early empirical work has documented positive outcomes. Rello and Baeza-Yates [3] found that automatic text simplification improved reading comprehension for dyslexic university students, particularly for informational texts. Stajner and Saggion [4] extended these findings to neural simplification models, noting gains in reading speed without significant loss of factual content. More recently, OpenAI et al. [5] reported that ChatGPT's conversational interface was perceived as helpful by students with ADHD for breaking down complex assignments. However, these studies evaluated simplification primarily on readability metrics (e.g., Flesch-Kincaid, SMOG) and user satisfaction, not on the accuracy of the simplified content when applied to technical domains.

Crucially, no study to date has examined whether LLM-generated simplifications for students with learning difficulties introduce novel errors that are absent in standard explanations. This gap is non-trivial: the cognitive mechanisms that make simplification beneficial, such as reduced decoding effort, shorter sentences, and concrete analogies, may simultaneously suppress the nuanced qualifications and precise technical language that prevent conceptual misunderstanding.

2.2. The Warmth-Accuracy Trade-Off and Accessibility-Driven Error

Recent work has established that LLMs exhibit systematic trade-offs between user-facing qualities and factual accuracy. Ibrahim et al. [12] demonstrated that fine-tuning models to produce warmer, more empathetic responses increases error rates across architectural families, with warm models showing a statistically significant tendency toward sycophantic validation of incorrect user beliefs. This trade-off is not an implementation flaw but a consequence of alignment pressure: models optimised for human preference judgments learn that users respond positively to agreeable, confident-sounding explanations, even when those explanations are technically incomplete.

Accessibility-driven simplification represents a distinct but related pressure. Where warmth optimisation prioritises affective alignment, simplification prioritises cognitive accessibility. The two are not identical: a simplified explanation need not be warm, and a warm explanation need not be simple. However, they share a common mechanism: both pressures push models away from precise, qualified, context-dependent technical statements and toward concrete, declarative, analogy-laden statements that are easier to process but more likely to be wrong in subtle ways.

We propose the term *oversimplification error* (OER) to capture a specific subclass of simplification-induced inaccuracies: errors that would not appear in a standard, technically precise explanation of the same concept but emerge exclusively as a product of the simplification process. OERs are distinct from random factual errors, such as misstating the year of an algorithm's invention, in that they arise systematically from the interaction between the concept's inherent complexity and the pressure to reduce that complexity. A standard explanation of semaphores, for example, would specify atomicity, counting semantics, and the distinction between blocking and busy-waiting. A simplified explanation that omits atomicity is not merely incomplete; it actively misleads by suggesting that semaphores are just integer variables with special names.

2.3. Text Simplification: Pedagogical Benefits and Technical Risks

Automatic text simplification has a long history in natural language processing, evolving from rule-based systems that applied syntactic transformations and lexical substitution to modern neural approaches using encoder-decoder architectures and, more recently, LLMs [4]. Evaluation has traditionally focused on three dimensions: readability (mea-

sured by standard formulas), semantic preservation (measured by entailment or meaning similarity metrics), and grammaticality (measured by fluency judgments).

This evaluation paradigm assumes a monotonic relationship between simplification and information retention: simpler texts should preserve the truth conditions of the original while reducing linguistic complexity. For many domains, including news articles, consumer health information, and government documents, this assumption holds reasonably well. The original and simplified versions describe the same facts using different words.

For computer science education, this assumption breaks down. CS concepts are not facts about the world but abstract, relational structures with precise definitions and boundary conditions. A linked list is not merely a chain of connected items, as that description also fits a rope, a queue, or a biological pathway. The pedagogical challenge is to convey both the intuitive metaphor and the precise conditions under which the metaphor fails. Good CS instruction does not replace technical precision with analogy; it layers analogy on top of precision, using metaphor as a scaffold, not a substitute.

LLMs optimised for simplification lack this pedagogical awareness. Under pressure to produce simpler output, they systematically omit boundary conditions, elide exceptions, and promote analogies from the intuitive level to the definitional level. The result is not a simplified version of the same concept but a different, incorrect concept that happens to share surface features with the target. A student who learns that a deadlock is two people stuck in a doorway has not learned a simplified but essentially correct version of deadlock; they have learned a misconception that will later need to be unlearned.

2.4. *Misconceptions in Computer Science Education*

The study of misconceptions in CS education is well established. Qian and Lehman [8] provide a comprehensive review, identifying common incorrect mental models across introductory programming, such as variable assignment as mathematical equality, the countdown misconception in loops, confusion between reference and value semantics, as well as data structures, such as believing that binary search trees are inherently balanced, and concurrency, such as thinking that race conditions are deterministic or that semaphores prevent all interleavings.

Crucially, prior work has shown that misconceptions are resistant to correction and that intuitive but incorrect analogies are a primary source of persistent misunderstanding [7]. Once a student has internalised a wrong mental model that successfully explains some subset of observed behaviour, they are unlikely to abandon it unless confronted with clear counterexamples that their model cannot handle. This makes the initial presentation critical: first exposure to a concept has outsize influence on final understanding.

Stone et al. [11] recently extended this analysis to LLM-generated educational content, documenting that common analogies produced by GPT-4 for CS concepts frequently misrepresent technical properties. However, their study used standard prompting without explicit accessibility constraints and did not compare simplification conditions. Our work builds directly on Stone et al. [11] by systematically varying simplification pressure and measuring the resulting amplification of strategic misconceptions.

2.5. *The Accessibility Paradox: A Formal Hypothesis*

Synthesising these three streams, we formalise the accessibility paradox as a testable hypothesis. For a given concept c in computer science, let $E_s(c)$ be an LLM-generated explanation under a standard prompt (baseline) and $E_a(c)$ be an explanation under an accessibility prompt requesting simplification for a student with reading or attention difficulties. The strategic misconception rate $\text{SMR}(E)$ is the probability that E contains a conceptually coherent but fundamentally incorrect understanding of c . The oversimplifica-

tion error rate $OER(E)$ is the probability that E contains an error present in E_a but absent from E_s for the same concept.

The accessibility paradox hypothesis states:

$$SMR(E_a) > SMR(E_s) \quad \text{and} \quad OER(E_a) > 0, \tag{1}$$

with the difference increasing as a function of simplification intensity. That is, making explanations more accessible for students with learning difficulties increases, rather than decreases, the rate of strategic misconceptions, and introduces a new class of errors that would not occur in standard explanations.

This hypothesis does not imply that simplification is inherently harmful or that LLMs should not be used as assistive technology. Rather, it predicts a previously undocumented failure mode that current evaluation practices do not detect. The AccessTrap-CS benchmark described in Section 3 was designed to test this hypothesis across models, subfields, and simplification intensities.

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Abbreviations

The following abbreviations are used in this manuscript:

LLM	Large Language Model
CS	Computer Science
SMR	Strategic Misconception Rate
DAR	Dangerous Analogy Rate
OER	Oversimplification Error Rate
ARS	Accessibility Risk Score
ADHD	Attention Deficit Hyperactivity Disorder
ANOVA	Analysis of Variance
HSD	Honestly Significant Difference
HPC	High Performance Computing

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